ASHIKUR RAHMAN

AI & Full-Stack Developer | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Versatile **AI & Full-Stack Developer** with hands-on software company experience and expertise in building scalable, intelligent web applications and computer vision systems. Skilled in integrating Machine Learning models (YOLO, CNNs, Transformers) into full-stack web platforms using React, Next.js, and Python backends. Experienced in low-latency model inference, RESTful API design, and hardware-software sensor integration. Focused on building robust software solutions, MLOps best practices, and explainable AI (XAI).

TECHNICAL CAPABILITIES

AI & ML Stack:	PyTorch, YOLOv8/v11, CNNs, BERT, SHAP (Explainable AI), OpenCV, Scikit-Learn, Pandas, NumPy
Full-Stack Web:	React, Next.js, TypeScript, JavaScript (ES6+), Tailwind CSS, Python (Flask), ASP.NET Core, REST APIs
Cloud & Databases:	Firebase, PostgreSQL, SQL Server, Docker basics, Cloud Platforms (AWS/GCP - Continuous Learning)
Tools & Workflows:	Git/GitHub, VS Code, Postman, Linux/Bash, Data Annotation Frameworks, Agile/Scrum

PROFESSIONAL EXPERIENCE

Full-Stack & AI Developer | Software & IT Solutions Firm
05/2024 – Present (Mohammadpur, Dhaka)

- Developing and maintaining client-facing full-stack web platforms using **React, Next.js, TypeScript, and Flask/Node backends**.
- Integrating machine learning capabilities and automated analytics dashboards into existing enterprise web applications.
- Collaborating with cross-functional engineering teams in Mohammadpur to optimize database queries and API response times.
- Implementing responsive UI components with **Tailwind CSS** and adhering to clean architectural practices.

Frontend & AI Developer | Freelance / Self-Employed
01/2022 – Present

- Architected and deployed responsive web apps using **React, Next.js, and Firebase** for international clients.
- Built reusable component libraries and performance-optimized interfaces, achieving top SEO and core web vitals scores.
- Maintained clear communication and delivered high-quality software projects on strict client deadlines.

AI Research & Model Developer | Academic & Independent Projects
2024 – Present

- Engineered real-time computer vision models using **YOLO and CNN architectures** for object detection and visual processing.
- Applied **SHAP (Explainable AI)** to interpret neural network outputs in robotics and automated classification tasks.

FLAGSHIP CAPSTONE PROJECT

Fire-Fighting Humanoid Robot (Capstone Lead)
2024 – 2025

Robotics | Edge AI | Computer Vision | Microcontrollers | Sensor Fusion

Designed and built an autonomous humanoid robot equipped with real-time AI fire detection and emergency response mechanics. Integrated computer vision model streams with spatial hardware sensors to perform dynamic hazard navigation, heat tracking, and immediate extinguishing actions.

KEY ENGINEERING PROJECTS

VisioSLR — Real-Time Sign Language Recognition System — 2024

Python | YOLOv8 | OpenCV | CNN | PyTorch

Built an end-to-end vision translation pipeline that converts live sign language gestures into text with high inference accuracy. Optimized frame processing rates using custom dataset augmentations and lightweight network layers.

Taskforce AI — Automated Business Agent Platform — 2024

Next.js | TypeScript | Tailwind CSS | RESTful APIs

Designed and implemented an agent platform automating business workflows, smart customer query routing, and automated response generation through backend API integrations.

CityPulse — Smart City Civic Analytics Platform — 2024

React | Chart.js | Flask | SQL Server

Architected a central dashboard unifying urban civic metrics, service complaint streams, and predictive municipal data into a single responsive web interface.

Personal Finance & Budget Analytics Tracker — 2023 – 2024

React | Firebase Auth & Firestore | Tailwind CSS

Engineered a secure single-page web application for tracking personal income, recurring expenses, and dynamic budget analytics graphs.

ACHIEVEMENTS & TECHNICAL MILESTONES

Capstone Hardware-AI Integration
Led the primary AI vision and sensor fusion integration for the Fire-Fighting Humanoid Robot at North South University.

Kaggle Data Science Benchmarks
Active participant in competitive image recognition and NLP challenges; authored reusable computer vision pipelines.

Language Certification
Earned official Chinese Language Certificate from the Confucius Institute at North South University.

Open Source & Client Delivery
Successfully completed multiple full-stack client builds with 100% on-time delivery across web and mobile platforms.

EDUCATION & CREDENTIALS

North South University — Dhaka, Bangladesh — 2020 – 2025

Bachelor of Science in Computer Science & Engineering (B.Sc. CSE)

Completed 100/134 Credits • Key coursework: Artificial Intelligence, Machine Learning, Operating Systems, Computer Networks, and Web Engineering.

HSC (Science) — 2019
Chittagong Port Authority College

SSC (Science) — 2017
Halishahar Cantonment Public School & College

CONTINUOUS LEARNING & FOCUS

- **MLOps & Cloud Deployment:** Containerizing ML microservices with Docker, FastAPI, and Kubernetes.
- **Generative AI & LLMs:** Exploring RAG architectures, prompt engineering, and vector database integrations.
- **Mobile & Security:** Cross-platform application development and web application security best practices.